…… Massive Crowd Management in Urban Pedestrian Zones

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***Abstract*—Open scenic areas and major event venues often face challenges such as intersecting pedestrian flows across multiple channels, rapid propagation of local congestion risks, and experience-based control schemes. This paper proposes a macroscopic crowd dynamics and optimization method for fine-grained large-flow management. Based on the Hughes continuum framework, the model couples pedestrian route choice with density conservation to describe multi-stage and multi-route crowd movements. Local admissible direction sets are used to represent rules such as one-way passage, two-way passage, and channel closure, while fixed anisotropic metrics describe geometric guidance induced by channels. Internal channel entrance capacity control is further introduced, enabling a control strategy to specify both permitted travel directions and time-varying release intensities. This paper constructs an evaluation and optimization framework integrating direction configuration, entrance capacity, actual channel load, entrance waiting, and control smoothness. A hierarchical constrained mixed-variable black-box optimization algorithm, HCMBO, is designed to jointly search direction and capacity strategies under fixed behavioral preferences and fixed geometric guidance parameters. Experiments on a simplified scenic-platform scenario show that the proposed macroscopic model can capture the complex interactions among multi-stage behavior, channel geometric guidance, and direction constraints. Under a composite objective weighted by total travel time, high-density exposure, actual channel-load imbalance, and related indicators, repeated experiments show that HCMBO achieves the lowest mean objective value of 2.6338, about 3.9% lower than the second-best TPE-style mixed Bayesian optimization method, and outperforms random search, pure simulated annealing, differential evolution, and other baselines. The proposed method provides a quantitative and interpretable decision-making tool for crowd routing and entrance flow control in open public spaces.**

***Index Terms*—Crowd dynamics, macroscopic modeling, channel optimization, Hamilton-Jacobi-Bellman equation, bi-level optimization.**

I. Introduction

LARGE-SCALE crowd gathering risks not only threaten individual safety but may also escalate into stampede incidents that disrupt social order and public security; the effectiveness of crowd management directly tests urban governance capacity and the modernization level of social safety governance systems [1], [2]. With the increasing frequency of high-density pedestrian activities such as holiday tourism, urban public events, commercial district promotions, New Year celebrations, and major sports events, core urban areas must accommodate pedestrian demands far exceeding daily levels within short time windows, posing enormous challenges for on-site organization, passage management, risk early warning, and emergency dispatch [3], [4], [5]. Against this backdrop, scientific modeling, numerical simulation, and optimization methods for formulating and evaluating crowd control strategies—improving the efficiency of large-scale event organization and reducing local congestion risks—have become critical scientific problems demanding urgent breakthroughs in social safety and urban governance [6], [7].

Crowds in large-scale events constitute typical complex systems. Numerous individuals with heterogeneous desired speeds, destinations, and route preferences interact through multi-scale, nonlinear mechanisms in confined spaces, generating spatiotemporal patterns at the collective level [8], [4]. Macroscopically, these patterns manifest as destination-oriented flow, passage convergence, bottleneck queuing, exit competition, and directional responses to control regulations [9], [10]; microscopically, individuals are influenced by surrounding density, visible paths, obstacle boundaries, and guidance facilities [11]. For open scenic areas, crowd behavior additionally exhibits pronounced phase dependence: visitors may first move toward the viewing platform, then tour along the main viewing direction, and subsequently choose different exit passages based on preferences [12]. These characteristics of multi-stage, multi-route coexistence with shared congestion mean that traditional static capacity analysis or single shortest-path models cannot adequately support refined management decisions.

Among various urban crowd gathering scenarios, open scenic areas, waterfront viewing platforms, and historic commercial districts are highly representative. Unlike enclosed venues such as stadiums and subway stations with well-defined boundaries [13], [14], such areas consist of open spaces, main tourist corridors, multiple lateral connecting passages, and return flow lines; visitors typically undergo a continuous behavioral chain of "entry–touring along the main corridor–choosing passages to leave–returning" [15], with crowd movement governed jointly by spatial geometry, control regulations, and historical behavioral preferences [16], [15]. Taking the Shanghai Nanjing East Road–Bund area as a concrete example, visitors approach the Bund from Nanjing East Road, ascend to the viewing platform via multiple stepped passages, tour along the waterfront, then exit through several southern passages and return along lower streets. This scenario can be abstracted as the typical "waterfront platform–multi-stepped passage" crowd flow scenario. In such scenarios, management difficulties manifest in three aspects: first, multiple passages simultaneously serve entry, exit, and diversion functions, and local direction settings can alter overall flow organization; second, distinct phase-dependent behavioral transitions exist between the main tourist corridor and lateral passages, making single origin–destination models insufficient; third, local congestion and global route choice form mutual feedback loops, where control changes in one passage may trigger load redistribution in adjacent areas. Fig. 1 shows the large-scale crowd at the Shanghai Nanjing East Road–Bund scenario

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Fig. 1. Large-scale crowd at the Shanghai Nanjing East Road–Bund scenario during a holiday period.

during a holiday period.

Existing large-scale event security management practices typically integrate system dynamics, system coupling, and system safety theory, combined with high-risk zone delineation and historical crowd behavior patterns, to formulate a series of control measures [17], [18]. These measures include deploying security personnel at critical locations, erecting temporary barriers for isolation and diversion, implementing one-way passage on key channels, and applying batch release in core areas [19], [20]. For instance, during major holidays, the Shanghai Bund area adopts batch release, passage direction regulation, south-entry north-exit routing, and reduction of head-on conflicts [21]; similarly, Nanluoguxiang main street in Beijing implements one-way passage measures (south entry, north exit) during peak periods [22], consistent in control mechanism with the Shanghai Bund approach.

However, current management measures for urban high-pedestrian-flow areas still rely considerably on manual experience and static contingency plans, with deficiencies remaining in scientific evaluation and refined optimization [23]. First, many control schemes are formulated based on historical experience, lacking quantitative characterization of the impact of passage direction settings, entrance release capacities, and phase-dependent behavioral preferences. Second, existing methods tend to focus on single-point statistics or local density monitoring, making it difficult to describe the dynamic coupling among "passage direction rules–entrance flow control–local optimal directions–collective density evolution–exit load distribution" [24]. Third, in open scenic area scenarios with multiple entrances, passages, and stages, neglecting visitor route preferences and phase transition behaviors can easily lead to misjudgment of passage loads and high-risk area locations. Finally, many existing simulation or optimization methods treat the crowd model as a black-box evaluator, failing to exploit the structural information among passage geometry, one-way rules, internal entrance capacities, and multi-route subpopulations [25], resulting in insufficient efficiency in control strategy search [26].

To address the above issues, this paper proposes a macroscopic crowd modeling and control optimization method incorporating passage constraints, fixed geometric guidance, multi-stage route behavior, and internal entrance capacity control, targeting the "waterfront platform–multi-stepped passage" open scenic area scenario. At the modeling level, this paper constructs a Bellman–conservation-law coupled crowd continuum model: different stage–route subpopulations possess independent potential functions with optimal directions computed via the discrete Bellman equation, while sharing the speed–density relationship determined by total density; one-way, bidirectional, and closed passage states are formulated as constraints on the local admissible direction set; and a fixed anisotropic metric tensor $M(x;\eta_0)$ characterizes geometric channel alignment. We further introduce internal entrance capacity variables $q_c^{\pm}(t)$ to distinguish attempted entrance flow from actually admitted flow, so that excessive demand remains as upstream waiting rather than disappearing from the system. At the optimization level, the controllable variable is $z = (s, q)$, whereas visitor route preferences $\hat{p}$ and the geometric guidance parameter η_0 are fixed model inputs. The scalarized objective combines total travel time, high-density exposure, realized channel-load imbalance, entrance waiting, and capacity-control smoothness. A hierarchical constrained mixed-variable black-box optimization framework (HCMBO) is designed to solve this direction-capacity problem under limited high-fidelity simulation budgets.

The main contributions of this paper are as follows.

1) A "waterfront platform–multi-stepped passage" crowd flow modeling framework for open scenic areas is proposed, representing visitor behavior as a multi-stage, multi-route continuum flow process, where different subpopulations possess independent potential functions while sharing collective congestion feedback.
2) A Bellman–conservation law coupled model integrating passage constraints and geometric guidance is constructed. The admissible direction set first encodes one-way, bidirectional, and closed channel states, while an anisotropic metric tensor simulates pre-alignment, smooth convergence, and streamline reorganization near passage entrances.
3) An evaluation framework distinguishing mechanism verification from strategy optimization is established, categorizing indicators into behavioral mechanism indicators (direction consistency, approach angle distribution, channel capture domain) and management optimization indicators (total travel time, high-density exposure time, realized channel-flow variance, entrance waiting, and control smoothness).
4) A hierarchical constrained mixed-variable black-box optimization framework is designed for the controllable direction-capacity variable $z = (s, q)$ under fixed historically identified preference parameters and fixed geometric guidance. The final HCMBO mainline removes auxiliary internal random search and concentrates the budget on direction-wise structured capacity optimization. The framework is evaluated against random search, pure simulated annealing, enumeration plus differential

evolution, and TPE-based mixed-variable Bayesian optimization.

II. Related Work

A. Massive Crowd Management and Control

Crowd management in urban centers has been extensively studied from control-theoretic and optimization perspectives, aiming to enhance safety and efficiency through boundary regulation, facility layout, and intelligent decision-making [27], [28].

In control-theoretic approaches, Zhong et al. [29] designed boundary control strategies for urban traffic flow networks to ensure convergence to uncongested states under disturbances, with stability analysis based on Lyapunov methods [30]. For one-dimensional crowd dynamics, Wadoo and Kachroo [31] employed advection and diffusion control to prevent blocking and shock waves during evacuation. Addressing multi-directional disturbance propagation, Qin [32] proposed a unit sliding mode controller based on integral barrier Lyapunov functions to ensure boundedness of internal state variables, demonstrating strong disturbance rejection capabilities. For heterogeneous corridor regulation, Zhu et al. [33] combined policy learning with neural networks to learn optimal feedback controllers that adjust inflow rates and free-flow speeds, achieving balanced density distribution and reduced congestion.

Physical infrastructure regulation has also proven effective. Studies have shown that strategically placed obstacles can significantly improve outflow, velocity, and local density in merging areas [23], [34]. Yang et al. [35], [36] employed Model Predictive Control (MPC) to optimize barrier lengths at subway bottlenecks, later extending control variables to entrance limiters, gates, and escalator directions for dynamic, real-time crowd regulation. For evacuation guidance, Carmona and Paricio-Garcia [37] proposed dynamic exit-choice recommendations using multinomial Logit models. However, their approach relies on discrete speed instructions (e.g., "stop," "walk," "run") transmitted via electronic wristbands [38], which poses implementation challenges in open urban scenarios.

Recently, intelligent optimization and deep reinforcement learning have emerged as promising alternatives. Liao et al. [39] formulated fence layout as a subset selection problem, while subsequent work [26] proposed a dynamic crowd inflow control system using LSTM networks to capture temporal features and Proximal Policy Optimization (PPO) [40] for real-time entrance flow regulation. Differential Evolution (DE) [41] has also been applied to optimize guardrail layouts and flow guidance in real-world scenarios such as Chengdu East Railway Station, using average travel time and crowd pressure as multi-objective criteria [42].

B. Macroscopic Crowd Modeling

Macroscopic crowd models treat pedestrian flows as continuous media, offering computational efficiency and analytical tractability compared to microscopic approaches. Henderson [43] first applied fluid dynamics to pedestrian traffic in the early 1970s, comparing crowd motion to the behavior of gas or fluid particles.

Hughes [44] pioneered the first-order continuum theory for pedestrian flow, introducing the classical model in which pedestrian walking speed is governed by local density through a fundamental diagram, while movement direction follows the steepest-descent path of a potential function representing the remaining travel time to exits. This framework elegantly captures the interplay between congestion and route choice. Ling et al. [45] subsequently generalized the Hughes formulation and developed an efficient numerical solution combining a weighted essentially non-oscillatory (WENO) scheme for the conservation law with a fast sweeping method for the eikonal equation.

Several extensions have enriched the descriptive capability of macroscopic models. Nonlocal flux models were proposed by Colombo et al. [46], replacing pointwise density with a neighborhood-averaged quantity in the velocity function to account for anticipatory behavior. This direction was further extended to incorporate anisotropic perception fields and boundary effects from walls and obstacles, with high-resolution numerical schemes and well-posedness results suitable for realistic architectural configurations [47]. To enforce the hard incompressibility constraint—that pedestrian density cannot exceed a physically admissible maximum—Maury et al. [48] introduced the projection model, which formulates the actual velocity as a least-squares projection of the desired velocity onto the set of feasible (non-overlapping) configurations. This framework was later recast as a Wasserstein gradient flow with density constraints [49], using the Jordan–Kinderlehrer–Otto (JKO) scheme [50] to discretize the motion while guaranteeing that density respects an upper bound, making it well-suited for high-density congestion scenarios.

To overcome non-physical behaviors such as supersonic information propagation that may arise in first-order models, Aw and Rascle [51] introduced a second-order model based on a generalized momentum conservation law that preserves anisotropy. The Aw–Rascle (AR) framework was subsequently adapted to pedestrian flows [52], where the congestion pressure term successfully reproduces characteristic phenomena including stop-and-go waves and spontaneous lane formation.

Beyond the classical Hughes-type framework, extensions have incorporated control and guidance mechanisms relevant to crowd management. Anisotropic formulations using metric tensors have been developed to represent channel geometries and directional preferences [53], [54], allowing for continuous geometric guidance without explicit prohibition. Constrained Hamilton-Jacobi-Bellman (HJB) formulations have been employed to model strict directional constraints such as one-way corridors, with the Bellman optimality principle providing a natural framework for unifying anisotropic geometric guidance with discrete directional constraints.

Despite these advances, existing approaches often treat control parameters as fixed or manually tuned. The joint optimization of discrete channel direction configurations and internal entrance capacity profiles remains underexplored, particularly for large-scale urban pedestrian zones with multiple behavioral stages and heterogeneous route choices.

III. Macroscopic Crowd Dynamics Model

A. Hughes Model Foundation

We consider a bounded two-dimensional domain $\Omega \subset \mathbb{R}^2$ representing the pedestrian walkway. Let $\rho(x,t)$ denote the crowd density at position x and time t, $\phi(x,t)$ the minimal remaining time to reach an exit, and $\mathbf{v}(x,t)$ the mean velocity field. The continuity equation governing mass conservation reads:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0. \tag{1}$$

In the classical Hughes model [44], pedestrians move in the direction of steepest descent of the potential function:

$$\mathbf{v} = f(\rho)\frac{-\nabla\phi}{|\nabla\phi|}, \tag{2}$$

where $f(\rho)$ is the speed-density relationship. We adopt the Greenshields form [55]:

$$f(\rho) = v_{\max}\left(1 - \frac{\rho}{\rho_{\max}}\right), \tag{3}$$

with $v_{\max}$ the free-flow speed and $\rho_{\max}$ the jam density.

The potential function satisfies the eikonal equation:

$$|\nabla\phi| = \frac{1}{f(\rho)}. \tag{4}$$

While this model captures basic congestion effects, it assumes that all directions are feasible and equally weighted. It therefore cannot represent one-way rules, temporary closures, or channel-induced geometric preferences. We address these limitations by first introducing HJB admissible-direction constraints for hard channel rules and then introducing the metric tensor to encode soft geometric guidance within the feasible direction set.

B. Channel Control via Constrained HJB

Many crowd management measures directly restrict available movement directions rather than merely adjusting traversal costs. Examples include one-way corridors, bidirectional passages, and temporary closures. Such rules are hard feasibility constraints on the local action set and are naturally modeled via constrained optimal control.

Let $U(x)$ denote the set of admissible unit directions at position x. The velocity satisfies:

$$\mathbf{v} = f(\rho)\mathbf{u}, \quad \mathbf{u}(x) \in U(x), \quad |\mathbf{u}| = 1. \tag{5}$$

By the Bellman optimality principle, the potential function satisfies the constrained HJB equation:

$$\max_{\mathbf{u}\in U(x)} \{-f(\rho)\mathbf{u}\cdot\nabla\phi\} = 1. \tag{6}$$

When $U(x)$ is the full unit circle, (6) reduces to the classical Hughes model. For restricted control sets, various channel rules can be expressed naturally. For channel c with region Ω_c and reference tangent $\tau_c(x)$, define

$$s_c \in \{-1, 0, +1, \varnothing\}, \tag{7}$$

where $+1$, -1, 0, and $\varnothing$ denote forward one-way, reverse one-way, bidirectional, and closed states, respectively. The corresponding admissible set is

$$U_c(x; s_c) = \begin{cases} \{\tau_c(x)\}, & s_c = +1, \\ \{-\tau_c(x)\}, & s_c = -1, \\ \{\tau_c(x), -\tau_c(x)\}, & s_c = 0, \\ \varnothing, & s_c = \varnothing, \end{cases} \quad x \in \Omega_c. \tag{8}$$

For a closed channel, the continuous maximum over an empty set is not used as an ordinary HJB state. Numerically, this state is handled as no feasible control in the channel region, equivalently $\phi(x) = +\infty$ for unreachable cells or infinite Bellman step cost from those cells.

C. Channel Geometric Guidance via Metric Tensor

The admissible set $U(x)$ determines which directions are feasible, but it does not fully describe how pedestrians approach and align with channel geometry before entering a passage. In real scenes, pedestrians often pre-align with the channel axis and merge smoothly near entrances rather than abruptly changing direction at the channel boundary. Therefore, after defining hard HJB constraints, we introduce a metric tensor $\mathbf{M}(x)$ to encode soft geometric preference within the feasible direction set.

To represent anisotropic channel geometries such as corridors and barriers, we introduce a spatially varying symmetric positive-definite metric tensor $\mathbf{M}(x)$ that generalizes the classical eikonal equation to anisotropic form [53]:

$$\sqrt{\nabla\phi^\top \mathbf{M}(x)\nabla\phi} = \frac{1}{f(\rho)}. \tag{9}$$

Under this metric, the optimal direction is determined by the tensor-induced steepest descent:

$$\mathbf{v} = f(\rho)\frac{-\mathbf{M}(x)\nabla\phi}{\sqrt{\nabla\phi^\top \mathbf{M}(x)\nabla\phi}}. \tag{10}$$

For a channel region Ω_c with unit tangent $\tau(x)$ and unit normal $n(x)$, we construct the metric tensor as:

$$\mathbf{M}(x) = \alpha(x)\tau\tau^\top + \beta(x)nn^\top, \quad x \in \Omega_c, \tag{11}$$

where $\alpha(x)$ and $\beta(x)$ control the traversal weights in tangential and normal directions, respectively. In the discrete Bellman update used in this paper, the directional step cost is proportional to $1/\sqrt{\mathbf{u}^\top\mathbf{M}(x)\mathbf{u}}$. Hence setting $\alpha(x) \gg \beta(x)$ lowers the effective cost along the channel tangent and suppresses cross-channel deviations, achieving continuous geometric guidance without explicit prohibition. Outside channels, $\mathbf{M}(x) = \mathbf{I}$ recovers the isotropic case.

Remark: The metric tensor represents directional preference rather than hard feasibility. As long as costs remain finite, pedestrians may still deviate when congestion or detours require it. Thus $U(x)$ answers which directions are allowed, whereas $\mathbf{M}(x)$ answers which allowed directions better follow the channel geometry.

D. Unified Discrete Bellman Formulation

To unify anisotropic geometry with discrete constraints, we employ a discrete Bellman formulation. For grid point x and admissible direction $\mathbf{u} \in U(x)$, define the step cost [56]:

$$\mathrm{Cost}(x, \mathbf{u}) = \frac{\Delta x}{f(\rho(x))} \frac{1}{\sqrt{\mathbf{u}^\top \mathbf{M}(x) \mathbf{u}}}, \tag{12}$$

The potential satisfies:

$$\phi(x) = \min_{\mathbf{u} \in U(x)} \left[\phi(x + \Delta x\, \mathbf{u}) + \mathrm{Cost}(x, \mathbf{u}) \right]. \tag{13}$$

This formulation couples metric tensor $\mathbf{M}(x)$ for geometric guidance with control set $U(x)$ for directional constraints within a unified Bellman framework. The optimal direction is:

$$\mathbf{u}^*(x) = \arg \min_{\mathbf{u} \in U(x)} \left[\phi(x + \Delta x\, \mathbf{u}) + \mathrm{Cost}(x, \mathbf{u}) \right], \tag{14}$$

with velocity $\mathbf{v}(x) = f(\rho) \mathbf{u}^*(x)$.

E. Multi-Stage Multi-Route Model

Real-world scenarios often involve complex itineraries with multiple stages and route choices. We extend the density field to include stage and route indices, with fixed-probability splitting at decision boundaries.

1) Stage-Route Decomposition: Let the complete behavior chain consist of S stages, where stage s has R_s alternative routes. The total density decomposes as [57]:

$$\rho(x, t) = \sum_{s=1}^{S} \sum_{r=1}^{R_s} \rho_{s,r}(x, t), \tag{15}$$

where $\rho_{s,r}(x, t)$ is the density of sub-population in stage s following route r.

All sub-populations share the same speed function $f(\rho)$ determined by total density, ensuring collective congestion effects. Sub-populations differ in their potential functions, optimal directions, and exit targets.

2) Sub-Population Potentials and Velocities: For each sub-population (s, r), define admissible direction set $U_{s,r}(x)$, metric tensor $\mathbf{M}_{s,r}(x)$, and target regions. The potential $\phi_{s,r}(x)$ satisfies:

$$\begin{aligned} \phi_{s,r}(x) = \min_{\mathbf{u} \in U_{s,r}(x)} \Big[& \phi_{s,r}(x + \Delta x\, \mathbf{u}) \\ & + \frac{\Delta x}{f(\rho)} \frac{1}{\sqrt{\mathbf{u}^\top \mathbf{M}_{s,r}(x) \mathbf{u}}} \Big]. \end{aligned} \tag{16}$$

The optimal direction is:

$$\begin{aligned} \mathbf{u}^*_{s,r}(x) = \arg \min_{\mathbf{u} \in U_{s,r}(x)} \Big[& \phi_{s,r}(x + \Delta x\, \mathbf{u}) \\ & + \frac{\Delta x}{f(\rho)} \frac{1}{\sqrt{\mathbf{u}^\top \mathbf{M}_{s,r}(x) \mathbf{u}}} \Big], \end{aligned} \tag{17}$$

with velocity $\mathbf{v}_{s,r}(x) = f(\rho)\, \mathbf{u}^*_{s,r}(x)$.

3) Stage Transitions and Fixed-Probability Splitting: Sub-populations couple through stage transitions. Let $G_{s,r} \subset \Omega$ be the completion goal region for stage (s, r) with indicator $\chi_{G_{s,r}}(x)$. Upon reaching this region, pedestrians switch to next-stage routes with fixed probabilities.

Let $p_{(s,r) \to q}$ be the fixed probability of switching from (s, r) to $(s + 1, q)$, satisfying:

$$p_{(s,r) \to q} \geq 0, \qquad \sum_{q=1}^{R_{s+1}} p_{(s,r) \to q} = 1. \tag{18}$$

The transition term is:

$$Q_{(s,r) \to (s+1,q)}(x, t) = p_{(s,r) \to q}\, \kappa_{s,r}\, \chi_{G_{s,r}}(x)\, \rho_{s,r}(x, t), \tag{19}$$

where $\kappa_{s,r} > 0$ is the stage switching rate.

The density evolution for each sub-population includes advection and transition source terms:

$$\frac{\partial \rho_{s,r}}{\partial t} + \nabla \cdot (\rho_{s,r} \mathbf{v}_{s,r}) = Q^{\text{in}}_{s,r} - Q^{\text{out}}_{s,r}, \tag{20}$$

where $Q^{\text{in}}_{s,r}$ is inflow from the previous stage and $Q^{\text{out}}_{s,r}$ is outflow to the next stage.

F. Internal Entrance Capacity Control

In addition to direction rules, managers may need to restrict how many pedestrians are admitted into an open channel during each time period. For channel c, we introduce entrance capacity controls

$$q = \{q_c^+(t), q_c^-(t)\}_{c=1}^{C}, \tag{21}$$

where the superscripts $+$ and $-$ denote the reference forward and reverse entrances. These controls constrain internal interface fluxes; they are not external boundary inflows.

The direction state and entrance capacity must satisfy hard consistency constraints:

$$\begin{aligned} s_c = +1 &\Rightarrow q_c^-(t) = 0, \\ s_c = -1 &\Rightarrow q_c^+(t) = 0, \\ s_c = \varnothing &\Rightarrow q_c^+(t) = q_c^-(t) = 0, \\ s_c = 0 &\Rightarrow q_c^+(t) + q_c^-(t) \leq \bar{q}_c. \end{aligned} \tag{22}$$

Let $A_c^\pm(t)$ be the attempted entrance flux induced by the unconstrained finite-volume update. The actually admitted flux is

$$\hat{A}_c^\pm(t) = \min\{A_c^\pm(t), q_c^\pm(t)\}. \tag{23}$$

The crossing flux at the internal entrance is scaled by $\hat{A}_c^\pm(t)/A_c^\pm(t)$ when $A_c^\pm(t) > 0$. The blocked part remains upstream and is recorded as entrance waiting or queueing mass, preserving the mass balance of the conservation-law update.

IV. Numerical Solution

We discretize the domain using a uniform two-dimensional grid, with the control direction set represented as an 8-neighborhood basis with masking for regional restrictions.

Unified Bellman Solver. Rather than separate eikonal and HJB solvers, we use a unified discrete Bellman shortest-path update. For candidate direction $\mathbf{u}$, precompute geometric step factors from the local tensor, then combine with the speed function to form the complete step cost. A priority queue propagates potential values outward from exits until convergence. When the control set is unrestricted, this procedure is equivalent to monotone solution of the anisotropic eikonal equation; when controls are restricted, it naturally reduces to constrained HJB solution, covering all required scenarios within a unified framework.

Density Update. After obtaining converged potentials, recover velocity fields using (17), substitute into the continuity equation, and advance density explicitly using an upwind flux scheme:

$$\rho^{n+1} = \rho^n - \Delta t \, \nabla \cdot (\rho^n \mathbf{v}^n). \tag{24}$$

The time step satisfies the CFL stability condition.

Evaluation Oracle. For any feasible control configuration $z = (s, q)$, the simulation workflow constitutes a lower-level evaluation operator:

$$(\rho, \phi, \mathbf{v}, B, R) = \mathcal{S}(z; \hat{p}, \eta_0), \tag{25}$$

where $\hat{p}$ represents fixed route preference parameters, η_0 is the fixed geometric guidance parameter, B records entrance waiting, and R records realized cumulative channel throughput. This operator provides the foundation for subsequent control optimization.

V. Control Evaluation Metrics

We distinguish between behavior validation metrics and control evaluation metrics. Behavior validation metrics include direction consistency, approach-angle distributions, and channel capture domains. Control evaluation metrics assess policy impacts under fixed $\hat{p}$ and η_0.

A. Efficiency, Safety, and Balance

For simulation horizon $[0, T]$, the total travel time metric is:

$$J_1(z) = \int_0^T \int_{\Omega_w} \rho(x, t) \, dx \, dt. \tag{26}$$

The high-density exposure metric is:

$$J_2(z) = \int_0^T \int_{\Omega_w} \mathbf{1}_{\rho(x,t) > \rho_{\text{safe}}} \, dx \, dt. \tag{27}$$

Let $R_c(T) = \int_0^T (\hat{A}_c^+(t) + \hat{A}_c^-(t)) \, dt$ be the realized cumulative throughput of channel c. The load-balancing metric is:

$$J_5(z) = \mathrm{Var}(R_1(T), \ldots, R_C(T)). \tag{28}$$

This metric is based on realized throughput R_c, not directly on the capacity control $q_c^{\pm}(t)$.

B. Entrance Waiting and Control Smoothness

Entrance waiting is penalized by

$$J_B(z) = \int_0^T \sum_{c=1}^{C} \left(B_c^+(t) + B_c^-(t) \right) \, dt. \tag{29}$$

If explicit queues are not stored, the blocked demand $(A_c^{\pm} - \hat{A}_c^{\pm})_+$ can be used as an equivalent waiting proxy.

For piecewise-constant capacity profiles, smoothness is measured by

$$J_R(z) = \sum_{c,k} \left[(q_{c,k+1}^+ - q_{c,k}^+)^2 + (q_{c,k+1}^- - q_{c,k}^-)^2 \right]. \tag{30}$$

C. Dimensionless Normalization

All terms are normalized before weighting. Let $M_{\text{tot}} = M_{\text{init}} + M_{\text{in}}$ denote the total evaluated crowd mass, let $|\Omega_w|$ be the walkable area, and let $R_{\text{tot}} = \sum_{c=1}^{C} R_c(T)$. We define

$$\tilde{J}_1(z) = \frac{J_1(z)}{M_{\text{tot}} T}, \tag{31}$$

$$\tilde{J}_2(z) = \frac{J_2(z)}{|\Omega_w| T}, \tag{32}$$

and

$$\tilde{J}_5(z) = \frac{\mathrm{Var}(R_1, \ldots, R_C)}{R_{\text{tot}}^2 (C-1)/C^2} = \frac{C^2}{C-1} \mathrm{Var}(r_1, \ldots, r_C), \tag{33}$$

where $r_c = R_c / R_{\text{tot}}$. Waiting and smoothness terms are normalized by fixed reference scales, giving $\tilde{J}_B$ and $\tilde{J}_R$.

D. Scalarized Optimization Objective

Given weights $(\lambda_1, \lambda_2, \lambda_5, \lambda_B, \lambda_R)$, the scalarized objective is:

$$J(z) = \lambda_1 \tilde{J}_1(z) + \lambda_2 \tilde{J}_2(z) + \lambda_5 \tilde{J}_5(z) + \lambda_B \tilde{J}_B(z) + \lambda_R \tilde{J}_R(z), \tag{34}$$

with the optimization problem:

$$\min_{z \in \mathcal{Z}} J(z), \qquad z = (s, q). \tag{35}$$

Our algorithm solves this scalarized problem for fixed weights, rather than directly generating the full Pareto frontier. Approximate Pareto sets can be obtained by running with different weight combinations.

VI. Optimization Method

A. Control Variable Definition

The controllable management variable is $z = (s, q)$. The direction block is

$$s = (s_1, \ldots, s_C), \qquad s_c \in \{-1, 0, +1, \varnothing\}, \tag{36}$$

and the capacity block is $q = \{q_c^+(t), q_c^-(t)\}_{c=1}^{C}$. In implementation, q is represented by a low-dimensional vector x through a direction-aware mapping

$$q = T_s(x), \qquad x \in [0,1]^d. \tag{37}$$

The mapping applies direction masks, capacity upper bounds, closure constraints, and smoothness restrictions.

B. Hierarchical Constrained Mixed-Variable Optimization

Treating this as an unconstrained black-box optimization ignores the hierarchical structure between direction rules and entrance capacities. We use HCMBO, a hierarchical constrained mixed-variable black-box optimization framework with five core components:

1) direction-candidate generation and filtering;
2) direction-aware capacity parameterization $q = T_s(x)$;
3) constrained surrogate or finite-budget black-box search over x;
4) derivative-free local refinement for promising candidates;
5) uniform high-fidelity re-evaluation for final ranking.

C. Direction Candidate Filtering

The outer level generates a set $\mathcal{S}_{\text{cand}}$ of direction configurations and removes candidates that violate operational constraints, connectivity requirements, or the direction-capacity consistency in (22). The retained set provides interpretable discrete alternatives for the inner capacity optimization.

D. Capacity Search and High-Fidelity Recheck

For a fixed direction configuration s, the inner level optimizes x in $q = T_s(x)$. The search may use Bayesian optimization, evolutionary search, or another finite-budget derivative-free method, but all candidates are passed through the same feasibility-preserving map T_s. The final HCMBO mainline uses the whole optimization budget for this direction-wise structured capacity search and derivative-free local refinement. It does not include an auxiliary internal random-search phase; random search is used only as a separate comparison method. All finalists are re-evaluated under the same high-fidelity simulator, and the paper reports only high-fidelity recheck results.

E. Compared Methods

The horizontal comparison includes Random Search, Pure SA (Pure Simulated Annealing), Enum-DE (Enumeration plus Differential Evolution), TPE-Mixed BO (Tree-structured Parzen Estimator based mixed-variable Bayesian optimization), and HCMBO. Pure SA refers to simulated annealing and should not be confused with the obsolete structure-aware block search used in earlier drafts.

VII. Numerical Experiments

This section presents numerical experiments to validate the proposed model mechanisms, control responses, behavior layer, and direction-capacity optimization framework. The experiments use a four-channel scenario, which is a simplified abstraction of the Bund waterfront viewing-platform environment. One side of the platform contains the initial crowd region and entry boundary, the other side contains the viewing and exit region, and a central wall contains four representative passages: upper, middle, lower-middle, and lower passages. The experiments follow a progressive logic: local mechanism validation is first conducted on the four-channel setting; the analysis then extends to multi-channel, multi-stage direction responses, behavior-layer validation, and finally horizontal optimization comparison over $z = (s, q)$.

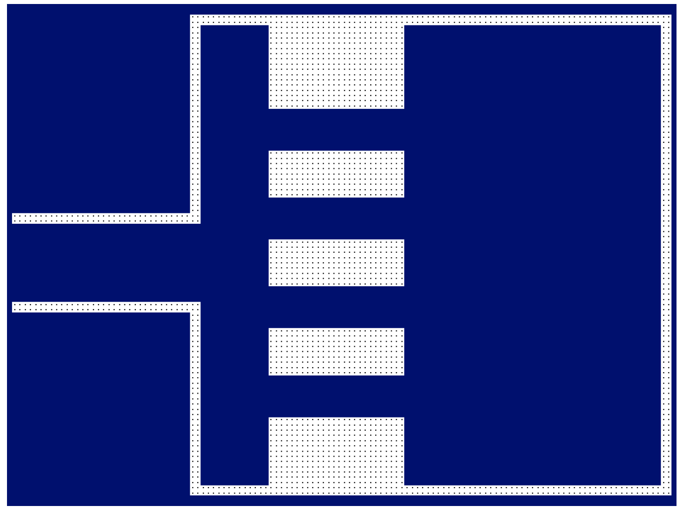

Fig. 2. Four-channel simulation scenario used in the numerical experiments.

A. Experimental Setup

1) Four-Channel Scenario: The four-channel scenic-platform scenario is used as the common basis for mechanism validation, strategy comparison, and optimization. The left side represents the initial crowd region and entry boundary, the right side represents the platform and exit target region, and the central wall contains four passages arranged from top to bottom as the upper, middle, lower-middle, and lower channels. Depending on the experiment group, pedestrians perform a single-stage crossing task, a bidirectional passage task, or a multi-stage “entry–tour–exit” behavior chain.

Numerical parameters are set as follows: grid resolution 128×96, spatial step $\Delta x = 0.5$, free-flow speed $v_{\max} = 1.5$, maximum density $\rho_{\max} = 5.0$, initial density $\rho_{\text{init}} = 2.2$, CFL coefficient 0.9, and safety density threshold $\rho_{\text{safe}} = 3.5$. The capacity-control horizontal comparison uses 1600 high-fidelity steps, a time horizon of $T = 160$, and Bellman updates every five simulation steps.

B. Mechanism Validation

1) Validation Logic and Experimental Matrix: The G1 mechanism validation experiment does not aim to rank control strategies. Instead, it verifies whether the two local mechanisms in the model have clear and interpretable behavioral meanings. The experiment is organized around four mechanism propositions, as shown in Table I.

This matrix separates the functional levels of $M(x)$ and $U(x)$. The former concerns preference changes within the feasible direction set, whereas the latter concerns modification of the feasible direction set itself. The final proposition then tests whether their combination produces synchronous migration when the controlled channel location changes.

TABLE I
G1 Mechanism Validation Matrix

Mechanism Proposition	Comparison	Main Observables	Expected Phenomenon
$M(x)$ represents soft geometric guidance	baseline vs. M-only	channel flow share; approach streamlines	target-channel attraction expands; inflow converges earlier
$U(x)$ represents hard directional constraints	baseline vs. U-only	restricted-channel flow; local direction field	restricted direction is excluded; flow shifts to feasible channels
$U(x)$ suppresses reverse flow and counterflow	bidirectional baseline vs. one-way rule	reverse mass time; head-on cell time	reverse flow and counterflow intensity decrease
$U+M$ is spatially transferable	different controlled-channel locations	dominant flow channel; hotspot location	dominant flow migrates with the controlled channel

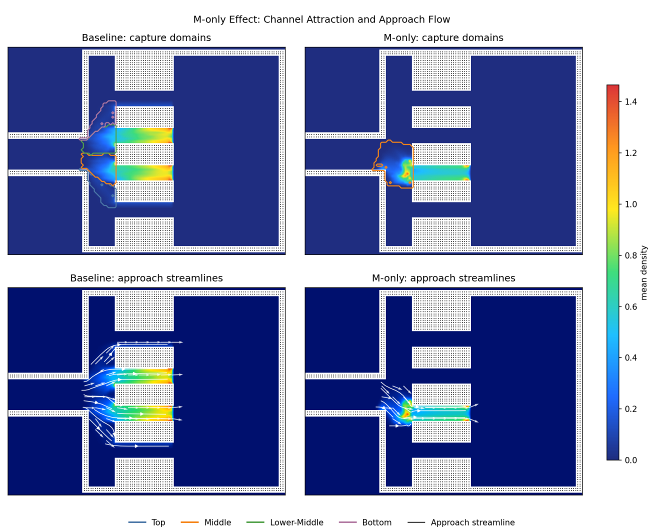

Fig. 3. Channel attraction domains and approach streamlines in the G1 experiment. The upper row shows capture-domain boundaries over the mean-density background, and the lower row shows time-averaged approach streamlines.

2) Soft Guidance Mechanism: $M(x)$ Alters Channel Attraction: To verify whether geometric guidance changes the approach process, we compare the baseline configuration against the M-only configuration, in which anisotropic geometric guidance is applied to the middle channel.

The baseline flow is mainly distributed between the middle and lower-middle channels: upper 1.64%, middle 48.78%, lower-middle 47.92%, and lower 1.66%. After introducing M-only guidance, the middle-channel flow share increases to 100.00%. As shown in Fig. 3, M-only guidance not only expands the attraction domain of the middle channel but also causes the incoming flow to converge toward the middle channel further upstream, forming a clear pre-alignment structure before the channel entrance. This indicates that the core role of $M(x)$ is to alter the relative travel cost within the feasible direction set, reshaping which channel attracts the crowd and when alignment begins, rather than directly prohibiting other passages.

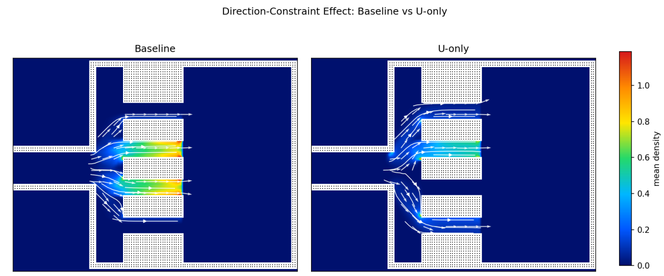

Fig. 4. Local direction field comparison between baseline (left) and U-only (right) configurations. White streamlines and arrows represent time-averaged movement directions; the background color encodes time-averaged density.

TABLE II
Bidirectional $U(x)$ Validation Results

Configuration	Westbound Share	Counter-flow Mass	Head-on Exposure
Bidirectional baseline	73.18%	415.75	1213.76
Middle channel one-way	0	0	0

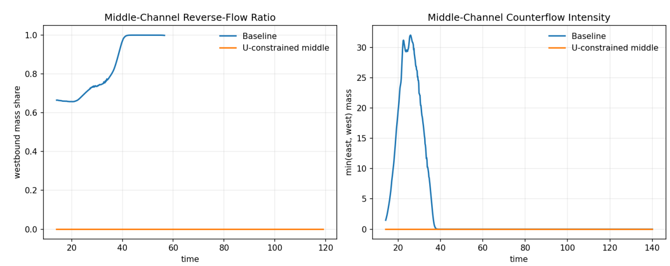

Fig. 5. Dynamic comparison of reverse flow in the middle channel under bidirectional conditions. Left: westbound mass share in the middle channel over time. Right: head-on conflict intensity in the middle channel over time.

3) Hard Directional Constraint Mechanism: $U(x)$: To verify whether directional constraints alter feasible path selection, we first compare the baseline configuration against the U-only configuration under unidirectional inflow, where the middle channel is subject to a directional constraint.

Under the U-only configuration, the middle-channel flow share drops to 0. The flow is redistributed to the upper, lower-middle, and lower channels, with shares of 19.25%, 68.76%, and 12.00%, respectively. This is the primary evidence for the role of $U(x)$: directional constraints do not continuously adjust route preference, but directly modify the local admissible direction set and thereby remove the restricted channel from the feasible choice set.

To further verify the ability of $U(x)$ to control directional conflicts, we conduct a bidirectional flow sub-experiment comparing a bidirectional baseline against a middle-channel eastbound one-way rule.

The bidirectional experiment shows that after imposing the

TABLE III
DOMINANT FLOW MIGRATION UNDER DIFFERENT $U + M$ CONTROLLED-CHANNEL LOCATIONS

Configuration	Dominant Flow Channel	Upper Flow	Middle Flow	Lower-Middle Flow	Lower Flow
baseline	middle/lower-middle	1.64%	48.78%	47.92%	1.66%
M-only-middle	middle	0	100.00%	0	0
$U + M$-middle	middle	0	100.00%	0	0
$U + M$-top	upper	100.00%	0	0	0
$U + M$-lower-middle	lower-middle	0	0	100.00%	0
$U + M$-lower	lower	0	0	0	100.00%

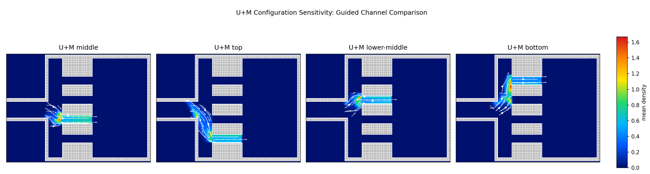

Fig. 6. Flow field comparison for the $U + M$ combined mechanism under different controlled-channel locations. The sub-figures correspond to different guided channels; white streamlines and arrows represent time-averaged movement directions, and the background color encodes time-averaged density.

one-way rule, westbound pedestrians are completely excluded from the middle channel, and both the middle-channel reverse-flow mass time and head-on cell time become zero. Therefore, $U(x)$ does not merely change path choice under unidirectional demand; it also persistently suppresses reverse flow and head-on conflict under bidirectional demand. This distinction clarifies the functional difference between $U(x)$ and $M(x)$: $M(x)$ adjusts route preference, whereas $U(x)$ modifies the feasible direction set. Thus, $U(x)$ is the hard-constraint representation of directional control policies.

4) Composability and Spatial Transferability of the Joint Mechanism: To verify the sensitivity of the joint mechanism to controlled-channel location, we test four $U + M$ configurations: middle-channel controlled, upper-channel controlled, lower-middle-channel controlled, and lower-channel controlled. The focus here is not on the efficiency or safety index of each configuration, but on whether the dominant flow channel changes synchronously when $U + M$ is applied to different channels.

The results show that changing the controlled-channel location induces synchronous migration of the dominant flow channel, density hotspot, and approach streamlines. When the upper, middle, lower-middle, or lower channel is selected as the joint guidance channel, the flow concentrates on the corresponding channel. This confirms that $U+M$ is not merely a local parameter setting that works at one fixed position; rather, it produces clear and interpretable behavioral responses as the spatial configuration changes. The joint mechanism can therefore serve as the lower-level simulator for subsequent strategy evaluation and optimization search.

C. System Response and Optimization Complexity Under Directional Control

After validating that $U(x)$ and $M(x)$ produce the expected local control responses, we further examine whether the discrete directional control variable s induces nontrivial system-level responses in the full multi-stage scenario. The purpose of this experiment is not to select a final optimal policy from a small enumeration, but to test whether interpretable perturbations of channel directions generate distinguishable changes in efficiency, safety exposure, and load balancing. Such evidence is needed before introducing an automated optimization procedure.

The complete combinatorial space of four-channel direction settings is large and contains many cases with weak operational meaning. We therefore construct a representative validation matrix with three policy families: single-entry (SE), single-return (SR), and one-channel-closed (CL), each rotated over the upper, middle, lower-middle, and bottom channels. Together with the all-free baseline (BSL), this gives $1 + 4 + 4 + 4 = 13$ cases. Direction codes are ordered as $[T, M, LM, B]$, where T, M, LM, and B denote the upper, middle, lower-middle, and bottom channels. The symbols E, W, F, and X denote eastbound entry, westbound return, free bidirectional passage, and closure, respectively. Case indices are retained only as configuration identifiers and are not mixed with policy-family labels.

Fig. 7 shows that different direction settings produce stable and clearly separated changes in $\tilde{J}_1$, $\tilde{J}_2$, and $\tilde{J}_5$. Using the aggregated normalized objective $\tilde{J} = \tilde{J}_1 + \tilde{J}_2 + \tilde{J}_5$, C8 (SR-LM, EEWE) obtains the lowest value, $\tilde{J} = 0.5102$. The baseline C1 is close, with $\tilde{J} = 0.5145$, but achieves the lowest safety exposure, $\tilde{J}_2 = 0.00785$. C9 (SR-B) gives the best load-balancing term, $\tilde{J}_5 = 0.1138$, while C3 (SE-M) gives the best efficiency term, $\tilde{J}_1 = 0.3141$, at the cost of higher safety exposure and load imbalance.

These results indicate that perturbing the direction variable s does not produce a monotonic improvement pattern, nor does any single configuration dominate all three objectives. The non-dominated set contains C1, C3, C4, C6, C8, and C9, confirming that the four-channel multi-stage scenario already forms a nontrivial multi-objective response structure.

The direct spatial mechanism behind these objective differences is the redistribution of channel loads and high-density hotspots. As shown in Fig. 8(A), the baseline C1 concentrates flow mainly in the middle and lower-middle channels, with shares of 34.0% and 47.2%, respectively. C8 has channel-load shares of 0.6%, 47.6%, 32.0%, and 19.7%, maintaining a low aggregate objective while still assigning the largest burden to the middle channel. C9 has load shares of 0, 30.0%, 33.7%, and 36.3%, which explains its best load-balancing term but weaker efficiency.

Fig. 8(B) further shows that directional control changes not only total flow allocation but also the spatial location of risk exposure. Hotspot locations are visualized in the true scene coordinate system and on the same feasible-domain and obstacle background as the density maps. Upper-channel-dominant or middle-closure configurations shift hotspots upward; middle-channel dominance places hotspots near the central passage; and lower-middle or bottom involvement shifts hotspots downward. Hence, the variations in $\tilde{J}_1$, $\tilde{J}_2$, and $\tilde{J}_5$ are not merely abstract numerical differences, but arise from coupled changes in channel-load redistribution, local safety exposure, and hotspot migration.

The G2 experiment therefore supports the need for subsequent optimization from three perspectives. First, the problem

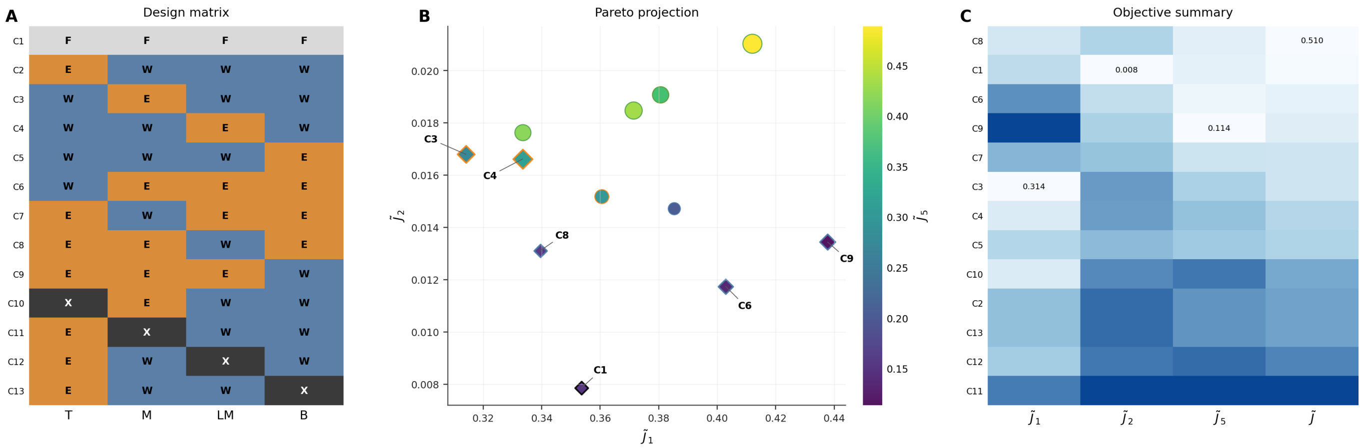

Fig. 7. Control-variable design and objective-space trade-off in G2. (A) Directional design matrix ordered by $[T, M, LM, B]$. (B) Pareto projection, where color encodes $\tilde{J}_5$, marker size encodes the aggregated normalized objective $\tilde{J}$, and diamond markers indicate non-dominated solutions. (C) Objective summary heatmap, with only the best value in each objective dimension annotated.

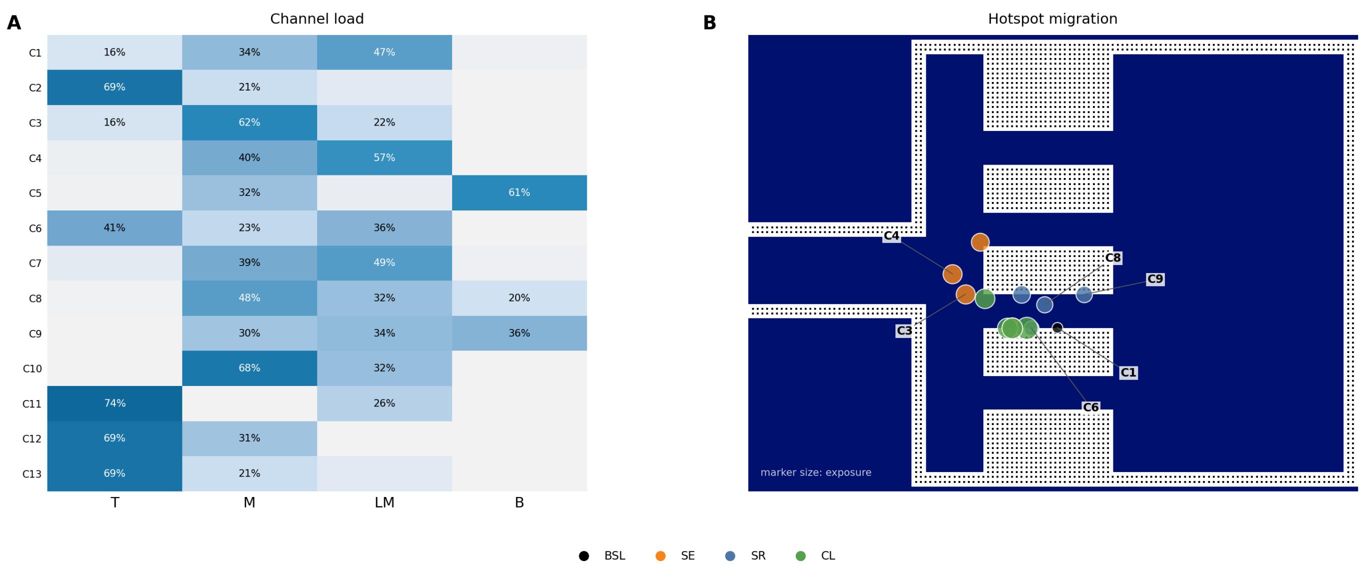

Fig. 8. Spatial mechanism behind directional-control responses. (A) Channel-load distribution across the 13 representative cases, with annotations retained only for major loaded channels. (B) Hotspot migration over the same scene background used in the density maps. Marker size represents local exposure intensity, and color indicates the policy family. If an exposure-weighted centroid falls inside an obstacle due to averaging over separated high-density regions, it is mapped to the nearest walkable cell for visualization.

TABLE IV
NON-DOMINATED DIRECTIONAL-CONTROL CASES IN G2

Case	Policy	Code	$\tilde{J}_1$	$\tilde{J}_2$	$\tilde{J}_5$	$\tilde{J}$	Main characteristic
C1	BSL	FFFF	0.3535	**0.00785**	0.1531	0.5145	Lowest exposure; close to the best aggregate objective
C3	SE-M	WEWW	**0.3141**	0.01680	0.2741	0.6049	Best efficiency; risk concentrates near the middle channel
C4	SE-LM	WWEW	0.3335	0.01662	0.3131	0.6632	Lower-middle entry; efficient but load-imbalanced
C6	SR-T	WEEE	0.4029	0.01174	0.1335	0.5482	Highest exit throughput; relatively balanced
C8	SR-LM	EEWE	0.3395	0.01311	0.1576	**0.5102**	Lowest aggregate objective
C9	SR-B	EEEW	0.4377	0.01344	**0.1138**	0.5649	Most balanced channel loads

is intrinsically multi-objective: efficiency, safety, and load balancing do not share a consistent optimum. Second, the direction variable is combinatorial, and the 13 cases here are only an interpretable slice rather than an exhaustive enumeration. Third, the full control problem also includes the continuous internal entrance capacity profile q. The resulting problem is therefore a mixed discrete-continuous, simulation-driven optimization problem with expensive evaluations, motivating the automated search procedure introduced later.

TABLE V
HIGH-FIDELITY BEST RESULTS IN THE DIRECTION-CAPACITY OPTIMIZATION COMPARISON

Method	Mean objective	Std.	Best	Worst	Feasible rate	Best seed
baseline_prior_best	5.4774	0.0000	5.4774	5.4774	0%	11
random_search	3.0112	0.1319	2.8821	3.2287	40%	51
pure_sa	2.9248	0.1386	2.7468	3.1049	60%	73
enum_de	3.0597	0.1932	2.7329	3.2500	40%	23
hcmbo_proposed	2.8848	0.1491	2.7002	3.0427	80%	37
tpe_mixed_bo	**2.7403**	0.2475	**2.5180**	3.1345	80%	23

TABLE VI
FOCUSED COMPARISON AFTER THE HCMBO MAINLINE UPDATE

Method	Best objective	Feasible rate	J_2	Gate rejected
HCMBO	**2.4495**	100%	**1.6759**	2545.508
TPE-Mixed BO	2.5180	100%	1.6948	2815.208
HCMBO queue-aware	2.6162	100%	1.8521	**2412.404**

D. Behavior-Layer Necessity

The behavior-layer experiment compares a single-stage approximation, a multi-stage model with uniform preferences, and a multi-stage model with heterogeneous fixed preferences. The single-stage approximation over-directs pedestrians to the middle channel, while the multi-stage models shift exit demand mainly toward the lower-middle and lower channels. The comparison shows that omitting the stage chain and route preferences can distort channel-load assessment and policy ranking. Therefore, $\hat{p}$ is treated as a fixed exogenous behavior parameter in the optimization experiment.

E. Horizontal Optimization Comparison

The completed horizontal comparison evaluates baseline prior best, Random Search, Pure SA, Enum-DE, TPE-Mixed BO, and HCMBO under the same high-fidelity setting. The result root is `codes/results/g6_horizontal_comparison`. Five seeds are used: 11, 23, 37, 51, and 73.

Table V shows that the G6 implementation of HCMBO outperforms Random Search, Pure SA, and Enum-DE in mean objective, and shares the highest feasible rate of 80% with TPE-Mixed BO. However, TPE-Mixed BO obtains the lowest mean objective and the best single-seed objective in this five-seed table. A post-hoc source audit further showed that the useful HCMBO candidates mainly came from direction-wise structured capacity search rather than auxiliary internal random search. Therefore, the final HCMBO mainline removes the internal random-search phase.

Table VI reports the focused comparison after this mainline update. Under seed 23, budget $B = 400$, top-10 high-fidelity rechecks, 1600 simulation steps, and time horizon 160, the updated HCMBO obtains a lower high-fidelity objective than the current TPE-style mixed BO baseline. Since this focused comparison uses one representative seed, it is used as direct evidence for the mainline update rather than as a claim of multi-seed statistical dominance.

VIII. CONCLUSION

This paper presents a Bellman–conservation-law crowd management framework for open pedestrian zones with channel direction configuration and internal entrance capacity control. The current controllable variable is $z = (s, q)$; the geometric guidance parameter η_0 and route preference parameter $\hat{p}$ are fixed model inputs. The proposed HCMBO framework exploits the direction-capacity hierarchy, concentrates the optimization budget on structured capacity search rather than internal random search, and uses high-fidelity rechecks for final ranking.

The numerical experiments validate that: (1) fixed geometric guidance $M(x; \eta_0)$ and directional constraints $U(x; s)$ alter local crowd behaviors in interpretable ways; (2) directional configurations induce nontrivial efficiency–safety–balance trade-offs; (3) multi-stage behavior modeling is necessary for reliable channel-load assessment; and (4) HCMBO outperforms Random Search, Pure SA, and Enum-DE in the multi-seed horizontal comparison, while the updated HCMBO mainline obtains a lower high-fidelity objective than the current TPE-style mixed BO baseline in the focused seed-23 comparison.

Future work will extend to larger real-world geometries, more random seeds, stronger official TPE/SMAC baselines, online demand estimation, multi-weight Pareto analysis, and calibration against field observations.

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